
Reproducibility Practices Review

A synthetic versioning, lineage, and rerun checklist.

Reproducibility in this packet means that a reviewer can follow a fictional run from protocol to export to review decision. It is a documentation practice, not evidence that a real result will replicate elsewhere.

Run lineage checklist

Checkpoint	Record to retain	Reviewer signal
Protocol identity	OCW-BIO-017 version label	method intent is fixed
Run lineage	ALPHA-A24-R03 and timestamp	export remains attributable
Instrument context	SpectraNova SN-4 configuration note	readout context is visible
Exception register	short rationale and owner	changes are reviewable
Review close	2026-07-13 17:30 UTC	decision timing is explicit

Versioning principle

A version label is useful only when it travels with the decision it shaped. The fictional packet therefore keeps the protocol identifier, run lineage, and exception note together. Recreating a file from memory is not treated as equivalent to a documented handoff.

Rerun decision tree

A bounded response to a synthetic review concern.

If this is observed	First question	Synthetic next action
Missing protocol version	Can the intended method be identified?	hold review; restore lineage before rerun
Control exception	Is the exception localized and documented?	rerun affected control placement
Instrument context absent	Can readout conditions be reconstructed?	do not interpret; recover configuration note
Duplicate discrepancy	Does CV exceed the planned threshold?	review handling and rerun paired wells
Complete lineage and controls	Is the conclusion still bounded?	close internal review at 17:30 UTC

Conclusion

The synthetic rerun decision is not a punishment for an imperfect value; it is a response to missing context. A reproducible packet makes uncertainty visible, records its decision path, and preserves the distinction between operational evidence and a real research claim.